

CIRO

Crisis Intelligence & Response Orchestrator

Verified intelligence. Faster response.

A Pakistan-first emergency platform connecting citizens, AI, response teams and command centers in one trusted loop.

Hackathon submission package: pitch deck, project document and system diagrams

PROJECT OVERVIEW

Executive summary

CIRO turns fragmented emergency reports into a verified incident state that can coordinate teams, warn affected citizens and support command decisions.

Core promise: verify once, coordinate everyone - while keeping sensitive actions under human authority.

- **Pakistan-first**

Mobile and web experiences are designed for citizens, responders and command teams, with support planned for Urdu, English and Roman Urdu.

- **Qwen-powered**

Alibaba Cloud Model Studio structures reports, summarizes evidence and supports risk classification. The AI recommends; policy and people authorize.

- **Cloud-native**

API Gateway, Function Compute, EventBridge, ApsaraDB RDS, OSS and Simple Log Service form a scalable and auditable implementation path.

- **Satellite-assisted**

Earth-observation data improves wide-area flood, wildfire, landslide and damage awareness when fused with ground evidence.

- **Hackathon MVP**

Report -> Qwen triage -> confidence gate -> dispatcher review -> responder and citizen notifications.

- **Pilot direction**

A controlled 90-day Sialkot pilot can validate accuracy, speed, trust and integration needs.

PROBLEM

The coordination gap costs critical time

Emergency signals often arrive through disconnected channels and require manual reconciliation before action.

● Scattered inputs

Calls, photos, messages, GPS points, sensors and agency updates do not automatically converge into one incident.

● Slow verification

Teams spend time confirming location, severity, duplicates and evidence quality.

● Late warnings

Nearby teams and communities may receive incomplete or delayed information.

DESIGN RESPONSE

CIRO creates a shared operating picture. Every report becomes structured evidence; related reports are clustered; risk is scored; exceptions are reviewed; verified events trigger targeted operational and public communication.

CIRO is not an autonomous emergency authority. It is a verification and coordination layer for authorized human operations.

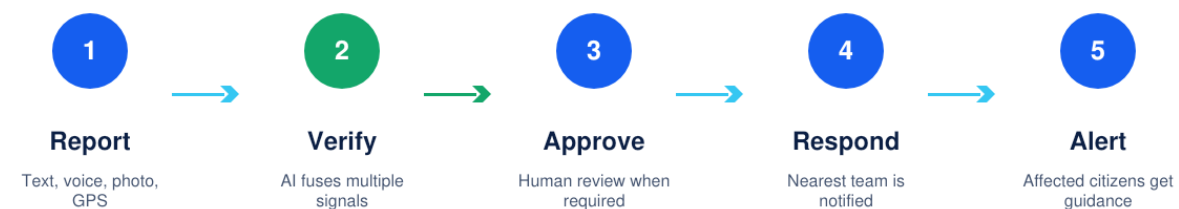
SOLUTION WORKFLOW

The verified response loop

A single, governed incident state replaces repeated manual handoffs.

CIRO creates one verified response loop

Every step is visible, governed and designed to reduce avoidable delay.



Design principle Automate detection and communication — keep authority, accountability and exceptions with people.

● Automation boundary

AI accelerates interpretation and routing. Sensitive dispatch, uncertain cases and public warning policies remain under configured authority.

PRODUCT ECOSYSTEM

Three connected products serve four roles

The same verified data is presented differently according to task and authorization.

● Citizen

Mobile app. Report with text, voice, photo and GPS; receive guidance; track status.

● Responder

Mobile/web workspace. Accept dispatch; navigate; review evidence; post scene updates.

● Dispatcher

Command dashboard. Review confidence; approve actions; coordinate units; manage alerts.

● Affected community

Alert channels. Receive geofenced, approved and incident-specific safety guidance.

Role-based access means citizens never see responder-only evidence, and public alerts exclude unnecessary personal information.

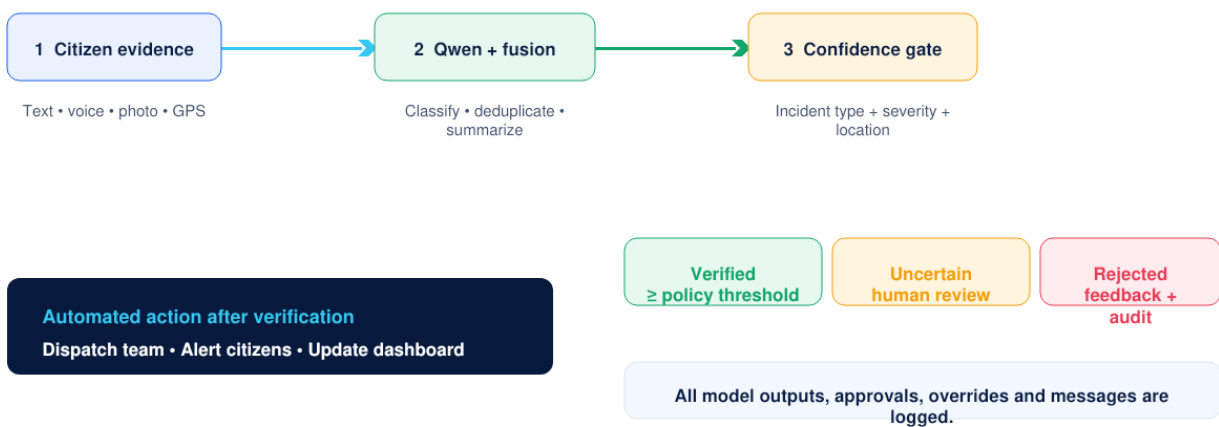
AI AUTOMATION

AI verification and alert workflow

Qwen structures multilingual evidence; confidence and policy determine the next safe step.

AI automates speed; policy controls what happens next

The workflow separates evidence processing from authority and public communication.



● Eligible automatic actions

Team pre-alerts, dashboard updates and template-based citizen notifications after verification policy is satisfied.

TECHNICAL ARCHITECTURE

Alibaba Cloud reference architecture

A modular service design supports rapid prototyping today and governed scale later.

CIROALIBABA CLOUD x BANO QABIL HACKATHON

CRISIS INTELLIGENCE & RESPONSE

10

Alibaba Cloud makes the response loop deployable

Serverless orchestration, managed AI, structured evidence and auditable events.

EXPERIENCE

Citizen app • Responder web • Command dashboard

INGEST + COMPUTE

API Gateway • Function Compute • EventBridge

AI LAYER

Model Studio / Qwen • PAI vision models • Policy engine

TRUSTED DATA

ApsaraDB RDS metadata • OSS evidence and imagery

GOVERNANCE

RAM + KMS access • Simple Log Service audit trail

Architecture references: Alibaba Cloud Model Studio, API Gateway, Function Compute, RDS/OSS and Simple Log Service documentation.

IMPLEMENTATION

Alibaba Cloud service mapping

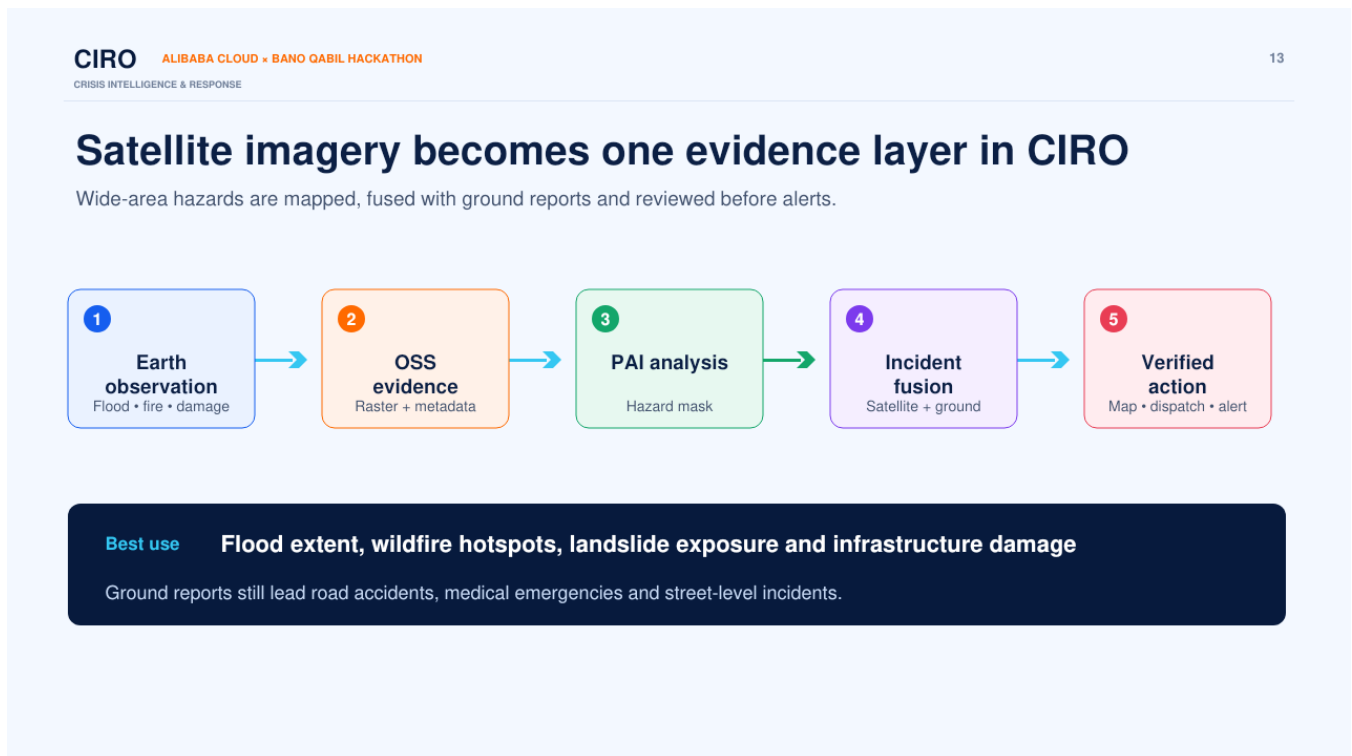
Each service has a clear responsibility and can be replaced or extended without redesigning the whole system.

SERVICE	ROLE	CIRO USE
API Gateway	Secure app and partner API entry	Authentication, throttling and stable endpoints
Function Compute	Event-driven orchestration	Runs intake, validation, routing and status workflows
Model Studio / Qwen	Language and multimodal intelligence	Classifies, summarizes and extracts incident fields
PAI	Vision and geospatial model serving	Processes hazard imagery and specialized classifiers
ApsaraDB RDS	Structured operational records	Incidents, users, teams, actions and status history
Object Storage Service	Evidence and raster storage	Photos, videos, documents and satellite imagery
EventBridge	Verified event routing	Decouples incident decisions from notification consumers
Simple Log Service	Operational and audit logs	Searchable record of outputs, approvals and delivery
RAM + KMS	Access and encryption controls	Least-privilege roles and managed keys

SATELLITE INTEGRATION

Satellite-assisted incident intelligence

Earth observation expands situational awareness for wide-area hazards; it does not replace immediate ground reporting.



- **Strong fit**
Flood extent, wildfire hotspots, landslide exposure and infrastructure damage assessment.
- **Ground-first**
Road accidents, medical emergencies and street-level incidents remain dependent on phones, responders and local sensors.

Candidate public sources: NASA FIRMS and Copernicus Emergency Management Service. Availability, latency and licensing must be validated per feed.

OPERATIONAL AUTOMATION

Automatic notifications after verification

CIRO can alert response teams and affected citizens in parallel, but with different content and authorization rules.

● Response teams

Dispatch request, incident type, priority, location, evidence summary, suggested route and acknowledgement status.

● Affected citizens

Geofenced safety alert, area to avoid, approved protective guidance, expiry time and official update channel.

● Command center

Live incident status, confidence, exceptions, team acceptance, notification delivery and complete audit trail.

● Message governance

Only approved templates are public. Private medical or personal details are never broadcast to the community.

Geofence logic follows the hazard: routes for road incidents, low-lying polygons for floods, and modeled spread zones for fires.

GOVERNANCE

Safety, privacy and responsible AI

Trust is designed into thresholds, data handling, access controls and auditability.

 Human authority

Sensitive and uncertain cases require dispatcher approval.

 Purpose limitation

Evidence and location are used only for the active safety workflow.

 Data minimization

Public alerts contain no unnecessary personal or reporter data.

 Approved guidance

Safety instructions use authority-reviewed templates.

 Audit by default

Model outputs, overrides and notifications are retained for review.

 Fail-safe behavior

Low confidence routes to review instead of automatic escalation.

 Role-based access

Citizens, responders and command staff see only what they need.

 Consent and retention

Permissions are explicit; evidence retention follows policy.

PRODUCT SPECIFICATION

Functional requirements

The MVP focuses on a complete, demonstrable incident lifecycle rather than a broad but shallow feature set.

● Report incident

Capture text/voice, media, GPS, category and consent.

● Structure evidence

Extract incident type, location, severity and entities.

● Detect duplicates

Cluster reports by time, place, similarity and media signals.

● Score confidence

Produce confidence, reasons, conflicts and recommended action.

● Human review

Approve, reject, merge, edit and override with required reason.

● Dispatch team

Select eligible nearby team and track acknowledgement.

● Alert citizens

Send geofenced, template-based guidance with expiry.

● Track timeline

Record status, model outputs, approvals and delivery events.

● Operate safely

Support access control, encryption, audit and deletion policy.

ENGINEERING OUTLINE

Core data model and API surface

A compact domain model keeps the prototype understandable and extensible.

● Incident

id, type, severity, location, confidence, state, timestamps

● Evidence

incident_id, source, media_uri, text, GPS, hash, quality

● Team

agency, capabilities, coverage, availability, last_location

● Decision

actor, model, action, reason, threshold, timestamp

● Notification

audience, geofence, template, channel, delivery state

● Consent

subject, purpose, permission, version, granted_at, revoked_at

CANDIDATE ENDPOINTS

**POST /reports GET /incidents/{id} POST /incidents/{id}/verify POST /dispatches POST /alerts
PATCH /incidents/{id}/status**

PROTOTYPE SCOPE

Hackathon MVP and demo flow

The demo proves the full response loop with simulated operational integrations.

01

Submit

Citizen creates a fire or flood report with photo, voice and GPS.

02

Analyze

Qwen extracts fields, summarizes evidence and returns confidence.

03

Verify

Related reports are clustered and a dispatcher approves the incident.

04

Respond

Nearest eligible team receives a dispatch request and accepts.

05

Alert

Affected citizens receive a geofenced, approved safety message.

06

Audit

Dashboard shows the incident timeline, decisions and delivery states.

External SMS, cell broadcast and government dispatch connections are represented by mock adapters until formal partners authorize integration.

DEPLOYMENT PLAN

A controlled 90-day Sialkot pilot

Start narrow, make every action observable, and expand only after evidence supports it.

● Days 0-30: Connect

Choose incident types, configure roles, create alert templates, establish consent and retention, and connect pilot responder groups.

● Days 31-60: Operate

Run controlled reports with dispatcher approval, limited geofences, simulated agency interfaces and daily incident review.

● Days 61-90: Evaluate

Measure accuracy, timing, false alerts, user trust, privacy events and technical integration needs.

PILOT PARTICIPANTS

● Citizen testers

Controlled participants and safety scenarios

● Dispatcher team

Review, approve and coordinate

● Response units

Receive and acknowledge dispatch

● Hospital / civic partner

Validate workflows and escalation

MEASUREMENT

Evaluation metrics and proposed targets

Targets guide validation; they are not claimed deployment results.

● **<= 60 sec**

Initial acknowledgement target for eligible digital reports.

● **<= 2 min**

Verified triage target for eligible pilot incident types.

● **100%**

AI decisions, approvals and overrides recorded in the audit trail.

● **Verification precision**

Correctly verified incidents / all verified incidents

● **False-alert rate**

Incorrect public alerts / all public alerts

● **Dispatch acceptance**

Time from verified event to team acknowledgement

● **Notification delivery**

Delivered alerts / intended recipients

● **Citizen trust**

Post-scenario clarity, confidence and consent ratings

● **Privacy incidents**

Unauthorized access, exposure or retention breaches

RISK MANAGEMENT

Key risks and mitigations

A crisis system must anticipate failure modes before real deployment.

False positive

Unnecessary dispatch or panic

Confidence gates, corroboration and approval policies

False negative

Missed or delayed response

Low-threshold pre-alerts and escalation monitoring

Malicious reports

Spam, manipulation or coordinated abuse

Identity signals, rate limits, hashes and audit review

Model uncertainty

Incorrect extraction or severity

Expose reasons, contradictions and human correction

Network outage

Reports or alerts cannot move

Offline queue, retry, SMS fallback and delivery state

Data exposure

Personal evidence reaches wrong audience

Least privilege, encryption, minimization and retention

Satellite latency

Old or cloud-obscured imagery

Show timestamp and quality; ground evidence remains primary

HACKATHON ALIGNMENT

Why CIRO fits the Alibaba Cloud x Bano Qabil Hackathon

The project combines meaningful AI, cloud engineering and responsible deployment around a nationally relevant challenge.

● Real-world impact

Emergency coordination is understandable, urgent and relevant across Pakistan's cities, campuses and industrial zones.

● Visible AI value

Qwen transforms unstructured multilingual reports into decisions the audience can see in the demo.

● Alibaba Cloud depth

The architecture uses managed AI, serverless compute, storage, events, database and observability services.

● Responsible design

Human oversight, privacy boundaries, confidence gates and auditability are part of the core workflow.

Submission story: CIRO is not only an AI model. It is a governed response system built around Qwen and Alibaba Cloud services.

NEXT STEPS

Roadmap and partnership ask

The next milestone is a controlled MVP demonstration followed by domain validation.

- **Hackathon**

Complete end-to-end fire/flood demo and cloud deployment.

- **0-3 months**

Validate with emergency-domain mentors and controlled users.

- **3-6 months**

Integrate real maps, notification providers and responder rosters.

- **6-12 months**

Run a governed pilot and prepare agency-grade controls.

REQUESTED SUPPORT

- Alibaba Cloud technical mentorship and credits
- Emergency-service domain feedback
- Controlled pilot access
- Mapping, satellite and notification partnerships

Every verified minute matters.

SOURCES

References

Primary sources used for event context, Alibaba Cloud architecture and external hazard feeds.

1. Bano Qabil AI Hackathon 2026 - <https://banoqabil.org/hackathon>
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3. Alibaba Cloud Global AI Hackathon Series - <https://www.alibabacloud.com/en/developer/global-ai-hackathon-series>
4. Model Studio overview - <https://www.alibabacloud.com/help/en/model-studio/what-is-model-studio>
5. Qwen API reference - <https://www.alibabacloud.com/help/en/model-studio/qwen-api-reference/>
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7. API Gateway supported backends - <https://www.alibabacloud.com/help/en/api-gateway/traditional-api-gateway/user-guide/support-for-multiple-api-backend-services/>
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9. EventBridge sources - <https://www.alibabacloud.com/help/en/eventbridge/user-guide/alibaba-cloud-service-event-sources>
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11. NASA FIRMS - <https://firms.modaps.eosdis.nasa.gov/>
12. Copernicus Rapid Mapping - <https://mapping.emergency.copernicus.eu/about/rapid-mapping-portfolio/>
13. Firebase topic messaging - <https://firebase.google.com/docs/cloud-messaging/topic-messaging>

Project metrics and scenario times in this document are proposed or illustrative. They are not claimed production results.